

Shrevatsan Venkatesan

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[LinkedIn](#) | [Portfolio](#)

Work experience

Staff Mechanical Engineer

Nov 2021 - Present

Clinical Diagnostics Group, Bio-Rad Laboratories Inc.

- Provide technical leadership for all Fluidics/Thermal/Mechanical design activities of new clinical diagnostics platforms, including architecture direction, procurement, and prototype creation.
- Evaluate and optimize design attributes and workflows for automated liquid handling robotics systems and microfluidics platforms, resulting in improved performance and efficiency.
- Collaborate with cross-functional teams to create and manage design requirements, specifications, FMEA, and V&V activities, ensuring compliance with industry regulations.
- Apply DFX/DFSS principles to develop cost-effective and reliable systems that meet customer needs and align with business strategies.
- Foster collaboration with Engineering disciplines, Chemistry, Regulatory, Marketing, Suppliers, and Manufacturing for seamless design transfer and successful product launch.
- Utilize CAD/CFD/FEA and analysis tools to manage designs and optimize them for fluidics, structural, thermal, and ergonomic requirements.
- Maintain clear and concise documentation of design processes, decisions and protocols facilitating effective communication and knowledge transfer within the team.

Sr. Product Development Engineer

Apr 2012 - Oct 2021

Vehicle Propulsion Systems Engineering, Ford Motor Company

- Played integral roles in various positions within the organization, contributing to the design, testing, and development of multiple vehicle platforms.
- Implemented a model-based systems engineering approach to optimize vehicle control and calibration strategies, resulting in improved vehicle attributes.
- Led the technical aspects of flow dynamic modeling for vehicle systems, combining real-world testing and virtual models to meet regulations and enhance performance.
- Developed innovative and robust air intake and exhaust system architectures through flow-thermal characterization and multi-criteria optimization.
- Utilized DFMEA, root cause analysis, DFSS, and 8D methodologies to identify and eliminate failure modes at various stages of vehicle development.
- Collaborated with global design and release groups to achieve targets for performance, NVH, cost, packaging, and manufacturability.
- Implemented design for manufacturability (DFM) best practices for plastic injection and blow molding technologies.

Automotive CAE Co-op

July 2011 - March 2012

Hitachi Automotive Products Research Laboratory, Michigan

- Flow modeling of vehicle fuel delivery systems.
- Model based design and validation of mass air flow sensors.

Research Intern

Dec 2008 - Apr 2009

Defense Electronics Research Laboratory, India

Design and Analysis of Cryogenic system for efficient treatment of satellite packaging material.

Education

Master of Science, Mechanical Engineering

Aug 2009 - July 2011

University at Buffalo, State University of New York

Bachelor of Technology, Mechanical Engineering

Jun 2005 - May 2009

Amrita Vishwa Vidyapeetham University, India

Patents/Publications

- 'Probe Cleaning System and Method' (Patent # WO2024191616A2)
- 'Manifold design for Active Droplet Generation on PCR chips'(Filed 2024)
- 'System for Decontamination of Instrument Wash Buffer Delivery System'(Filed 2025)
- 'Automated Reagent Pack Loading and Piercing for Bio-Plex Reagents'(Filed 2025)
- 'Separator Design for Segregation of Fluid/Gas Mixture'(Filed 2025)
- 'Sampling Tube Design for Sensor Robustness' (Patent # US20220235688A1)
- 'Convolute-Swirl Integrated duct for Swirl Generation' (Patent # US20190323460A1)
- 'Filter Element Robustness Strategy for Mud Ingestion' (SAE #2019-01-0916)
- 'CFD methodology to troubleshoot High Frequency Turbocharger NVH Error States' presented at 2018 Ford Global NVH Conference

Summary

- Experienced in Characterization, Development and Optimization of Electro-Fluidic-Mechanical Systems.
- Facilitated collaboration among cross functional teams to negotiate tradeoffs and troubleshoot failure modes.
- Leveraged virtual simulation and statistical tools to expedite development and reduce cost.
- Co-ordinated activities to build/test prototypes and identify potential suppliers.
- Received multiple Achievement Awards for developing product and process methodologies towards improving Performance and Reliability across multiple industries.

Area of Expertise

- CFD/FEA
- Multiphase Flow/Heat Transfer
- Injection Molding
- Microfluidics
- Six Sigma
- FMEA/Reliability
- DOE & Optimization
- Data Mining & Statistics
- Change Mgt./CAPA
- Design Control/ISO13485
- Robotics

Modeling/Simulation Skills

Physics Modeling and Visualization:

Ansys CFD/Mechanical, Comsol, Converge, STAR-CCM+, AFT Fathom, Simulink, Flomaster, ANSA Meshing

Data/Statistical Modeling:

Pandas,NumPy,SciPy, Minitab, Modefrontier, DesignXpert, PowerBI

CAD/PLM/PDM:

SOLIDWORKS(CAD/PDM/Cideon), SAP-PLM, Siemens Teamcenter, ProE, Spaceclaim

Programming : Python, Matlab, C++

Project Mgt :

JIRA, JAMA, Azure DevOps, Github, Veeva QMS

Professional

Courses/Certificates

- Design for Six Sigma Green Belt
- Lean Six Sigma
- Design Control Requirements (AAMI)
- Python Data structures(Coursera)
- Data Analysis with Python(IBM)
- Machine Learning with Python(IBM)
- Continuous Improvement(Bio-Rad)
- FMEA/Reliability(Ford)
- Ansys CAE/CFD Toolset
- Siemens CAE/CFD Toolset